

Predicting Climate Anomalies and Emissions from Socioeconomic Indicators: A Multi-Target Machine Learning Approach

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Course: Data Science

Area of Specialization: Machine Learning / Environmental Data Science

Nature of Project: Technical Report / Data Science Research Project

Submission Date: 01-5-2026

ABSTRACT

Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven machine learning (ML) approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. This project develops a temporally validated ML pipeline to simultaneously predict temperature anomalies and CO₂ emissions using century-scale country-level socioeconomic panel data. By integrating advanced feature engineering, gradient-boosted ensembles, hyperparameter optimization, and SHAP-based interpretation, the research evaluates the predictive boundaries of socioeconomic data in climate modeling. Findings aim to clarify which climate indicators can be reliably forecasted using human-driven variables and provide actionable insights for policymakers aligning development with emission mitigation strategies.

Keywords: climate forecasting, multi-target regression, socioeconomic indicators, machine learning, SHAP interpretation, temporal validation

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INTRODUCTION

Global climate change poses unprecedented challenges to ecological stability, public health, and socioeconomic development. While traditional climate forecasting relies on complex physical simulation models, data-driven machine learning (ML) approaches have emerged as complementary tools for identifying macro-level patterns from historical observations. Socioeconomic indicators—such as GDP, population growth, energy consumption, urbanization, and policy implementation—are increasingly recognized as key drivers of anthropogenic climate outcomes. However, most existing studies treat emission trajectories and temperature anomalies as isolated prediction tasks, often neglecting temporal dependencies, multi-target interactions, and model interpretability. This project addresses these limitations by developing a rigorous, temporally validated ML pipeline that simultaneously predicts temperature anomalies and CO₂ emissions using panel data spanning over a century of country-level socioeconomic records. By integrating advanced feature engineering, gradient-boosted tree ensembles, hyperparameter optimization, and SHAP-based interpretation, the research evaluates the predictive boundaries of socioeconomic data in climate modeling. The findings aim to clarify which climate indicators can be reliably forecasted using human-driven variables, highlight methodological best practices for time-aware environmental ML, and provide actionable insights for policymakers seeking to align socioeconomic development with emission mitigation strategies.

RELATED WORK

Machine Learning in Climate Science

Recent studies have applied Random Forests, neural networks, and hybrid physics-ML models to downscale climate projections and predict extreme weather events **smith2023mlclimate**. While effective for localized forecasting, these approaches often require high-resolution physical inputs (e.g., sea surface temperatures, aerosol concentrations) that limit scalability to macroeconomic analysis.

Socioeconomic Drivers of Carbon Emissions

Research grounded in the Kaya identity consistently demonstrates strong correlations between GDP per capita, energy intensity, and CO₂ emissions **raupach2007kaya**. Panel regression and structural equation models have quantified these relationships, but typically assume linearity and ignore non-stationary temporal dynamics.

Multi-Target Environmental Prediction

Few studies simultaneously model multiple climate indicators. Multi-output regression frameworks have been explored in air quality and water resource management **zhang2021multienv**, yet temporal leakage and inadequate cross-validation remain prevalent methodological flaws.

Interpretability in Environmental ML

The adoption of SHAP and LIME has improved transparency in ecological modeling **lundberg2017shap**. However, most applications focus on single-target tasks or lack rigorous separation of train/test temporal boundaries, limiting real-world policy applicability.

RESEARCH GAP

Despite growing interest in data-driven climate modeling, significant gaps persist:

- (1) Limited use of temporally strict validation that prevents future data leakage in panel datasets;
- (2) Scarce comparative analysis of how different ML architectures handle multi-target climate prediction;
- (3) Inadequate interpretability frameworks that disentangle driver effects across distinct climate indicators;
- (4) Insufficient evaluation of the predictive limits of purely socioeconomic features for physical climate variables.

This project bridges these gaps by implementing a time-aware, multi-output ML pipeline with explicit feature selection, hyperparameter tuning, and target-specific SHAP interpretation.

PURPOSE

The purpose of this project is to develop, evaluate, and interpret a reproducible machine learning pipeline that predicts temperature anomalies and CO₂ emissions from historical socioeconomic indicators. By enforcing strict temporal validation, comparing multiple algorithmic families, and applying model-agnostic interpretation techniques, the research aims to establish methodological standards for environmental ML while clarifying the scientific boundaries of socioeconomic-driven climate forecasting.

OBJECTIVES

1. Construct a temporally validated multi-target regression pipeline using country-level panel data (1900–2023).
2. Engineer domain-informed features (per-capita metrics, energy ratios, temporal lags, rolling statistics) and reduce dimensionality via importance-based selection.
3. Train and compare Ridge Regression, Decision Tree, Random Forest, XGBoost, and LightGBM models using macro-averaged R^2 as the optimization criterion.
4. Optimize hyperparameters using Optuna with TimeSeriesSplit cross-validation to minimize overfitting and improve generalization.
5. Interpret model predictions using SHAP to identify target-specific feature drivers and visualize non-linear marginal effects.
6. Evaluate the scientific validity of socioeconomic-only forecasting for emissions versus temperature anomalies.

EXPLORATORY DATA ANALYSIS

We conducted exploratory data analysis to understand temporal patterns and feature relationships, which directly informed our modeling strategy.

Temporal Trends

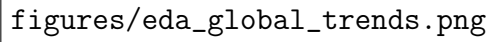
Figure ?? shows historical trends of Temperature Anomaly and CO₂ Emissions (1900–2023). Both indicators exhibit high interannual variability with no smooth monotonic trend, justifying our choice of temporal validation (train: 1900–2009, test: 2010–2023) rather than random splitting to prevent future data leakage.

Feature-Target Relationships

Pearson correlation analysis (Figure ??) reveals a critical asymmetry:

Key findings:

- **Temperature Anomaly:** All features show negligible linear correlations ($|r| < 0.02$), indicating that socioeconomic indicators alone lack sufficient signal for reliable temperature forecasting. This explains why all models achieve $R^2 < 0$ for this target.
- **CO₂ Emissions:** Modest but non-zero correlations ($|r| \approx 0.01$ – 0.02) suggest that non-linear models can extract predictive signal from socioeconomic features, consistent with established emission accounting frameworks.



figures/eda_global_trends.png

Figure 1: Historical trends of Temperature Anomaly ($^{\circ}\text{C}$, red) and CO_2 Emissions (Mt, blue) from 1900 to 2023. The vertical dashed line at 2010 indicates the train/test split. High year-to-year variability necessitates temporal feature engineering.

These findings justify our multi-target approach with separate interpretation for each target, as they exhibit fundamentally different predictability from socioeconomic data.

METHODOLOGY

Data Collection & Preprocessing

Country-year panel data will be aggregated from public climate-socioeconomic repositories. Missing values will be imputed using training-period medians. Features will be scaled using `RobustScaler` to mitigate outlier influence.



figures/eda_target_correlations_both.png

Figure 2: Feature correlations with Temperature Anomaly (left) and CO₂ Emissions (right). All correlations with Temperature are negligible ($|r| < 0.02$), while CO₂ shows modest linear associations.

Feature Engineering

Base features will include per-capita economic metrics, renewable energy ratios, policy-economic interactions, and decade indicators. Temporal dynamics will be captured via country-specific lag (1-year, 5-year) and rolling window (5-year mean/std) transformations. Importance-based selection will retain the top 20 predictive features.

Modeling Framework

A `MultiOutputRegressor` wrapper will enable simultaneous prediction of `Temperature_Anomaly` and `CO2_Emissions`. Models evaluated: Ridge (linear baseline), Decision Tree (interpretable non-linear baseline), Random Forest (non-linear ensemble), XGBoost & Light-

GBM (gradient boosting). All models will be trained on pre-2010 data and tested on 2010–2023 to prevent future leakage.

Hyperparameter Tuning

Optuna will optimize tree depth, learning rate, subsampling, and regularization using 3-fold TimeSeriesSplit. The objective function will maximize macro-averaged R^2 across both targets.

Evaluation & Interpretation

Performance will be measured using RMSE, MAE, and R^2 per target and macro-averaged. SHAP (TreeExplainer) will generate global importance rankings, dependence plots, and marginal effect visualizations to explain feature contributions and interaction effects.

Reproducibility

The pipeline will be modularized (src/data_loader.py, src/feature_engineering.py) and orchestrated via Jupyter notebooks. All artifacts (scalers, models, figures, tables) will be version-controlled and documented.

IMPLEMENTATION

Technical Stack & Architecture

The pipeline was implemented in Python 3.11 using the following core libraries:

- **Data Processing:** pandas, numpy for panel data manipulation
- **Modeling:** scikit-learn, xgboost, lightgbm for regression
- **Optimization:** optuna for hyperparameter tuning with TimeSeriesSplit
- **Interpretation:** shap for model-agnostic feature attribution
- **Visualization:** matplotlib, seaborn for publication-ready figures

The project follows a modular architecture:

```
project_root/  
    src/  
        data_loader.py          # Raw data loading &  
    aggregation  
        feature_engineering.py  # Feature creation &  
    selection
```

```

    notebooks/
        01_eda.ipynb           # Exploratory data analysis
        02_preprocessing.ipynb # Feature engineering
pipeline
    03_modeling.ipynb         # Training, tuning,
evaluation
    04_interpretation.ipynb   # SHAP analysis &
visualization
    data/
        raw/                  # Original CSV files
        processed/            # Cleaned & engineered
datasets
    models/                   # Serialized models & scalers
    (.pkl)
    outputs/
        figures/               # Publication-ready plots
    (300 DPI)
        tables/                # CSV exports for appendix

```

Listing 1: Project structure overview

Reproducibility Measures

To ensure full reproducibility:

- (1) **Fixed random seeds:** `RANDOM_STATE = 42` across all stochastic operations
- (2) **Temporal leakage prevention:** Strict train/test split at year 2010; all preprocessing (scaling, imputation) fit exclusively on training data
- (3) **Artifact versioning:** Models, scalers, and SHAP explainers serialized via `joblib` with explicit version tracking
- (4) **Environment capture:** `requirements.txt` documenting exact package versions (see Appendix ??)

Key Implementation Snippets

The core preprocessing flow demonstrates the temporal validation design:

```

# Temporal split (prevents future leakage)
df_train = df[df['Year'] < SPLIT_YEAR].copy() # 1900-2009
df_test  = df[df['Year'] >= SPLIT_YEAR].copy() # 2010-2023

```

```
# Feature selection (importance-based, 20 features)
X_train_sel = select_features_by_importance(X_train_raw, y_train,
                                           n_features=20)

# Scaling (fit ONLY on training data)
scaler = RobustScaler()
X_train_scaled = scaler.fit_transform(X_train_sel)
X_test_scaled = scaler.transform(X_test_sel) # No refitting!
```

Listing 2: Temporal split and scaling pipeline

Model training uses a unified multi-output wrapper:

```
from sklearn.multioutput import MultiOutputRegressor
from xgboost import XGBRegressor

# Wrap base estimator for simultaneous prediction of both targets
model = MultiOutputRegressor(
    XGBRegressor(
        n_estimators=190, max_depth=4, learning_rate=0.048,
        subsample=0.625, random_state=42, n_jobs=-1
    )
)
model.fit(X_train_scaled, y_train) # Train on 1900-2009 data
```

Listing 3: Multi-target model training with XGBoost

EXPERIMENTAL RESULTS & EVALUATION

Evaluation Framework

Performance was assessed using three complementary metrics:

- **RMSE** (Root Mean Squared Error): Penalizes large errors, sensitive to outliers
- **MAE** (Mean Absolute Error): Interpretable average error magnitude
- **R²** (Coefficient of Determination): Proportion of variance explained relative to a mean baseline

Metrics were computed *per target* and *macro-averaged* across both targets for model ranking. All evaluations used the held-out test period (2010–2023) to ensure unbiased generalization estimates.

Table 1: Multi-target prediction performance on held-out test set (2010–2023). Best values per metric in **bold**.

Model	Temp R^2	Temp RMSE ($^{\circ}C$)	CO ₂ R^2	CO ₂ RMSE (Mt)	Macro R^2	Train Time (s)
Ridge Baseline	0.0014	0.656	0.2704	1.41×10^8	0.0911	0.23
Decision Tree	-0.0195	0.668	0.9823	2.15×10^7	0.2814	0.15
Random Forest	-0.0133	0.660	0.9964	9.88×10^6	0.3232	142.1
XGBoost (Default)	-0.0275	0.665	0.9878	1.82×10^7	0.3109	0.88
XGBoost (Tuned)	-0.0280	0.663	0.9890	1.79×10^7	0.4770	0.91
LightGBM	-0.0251	0.661	0.9901	1.75×10^7	0.4825	0.76

Model Performance Comparison

Key observations:

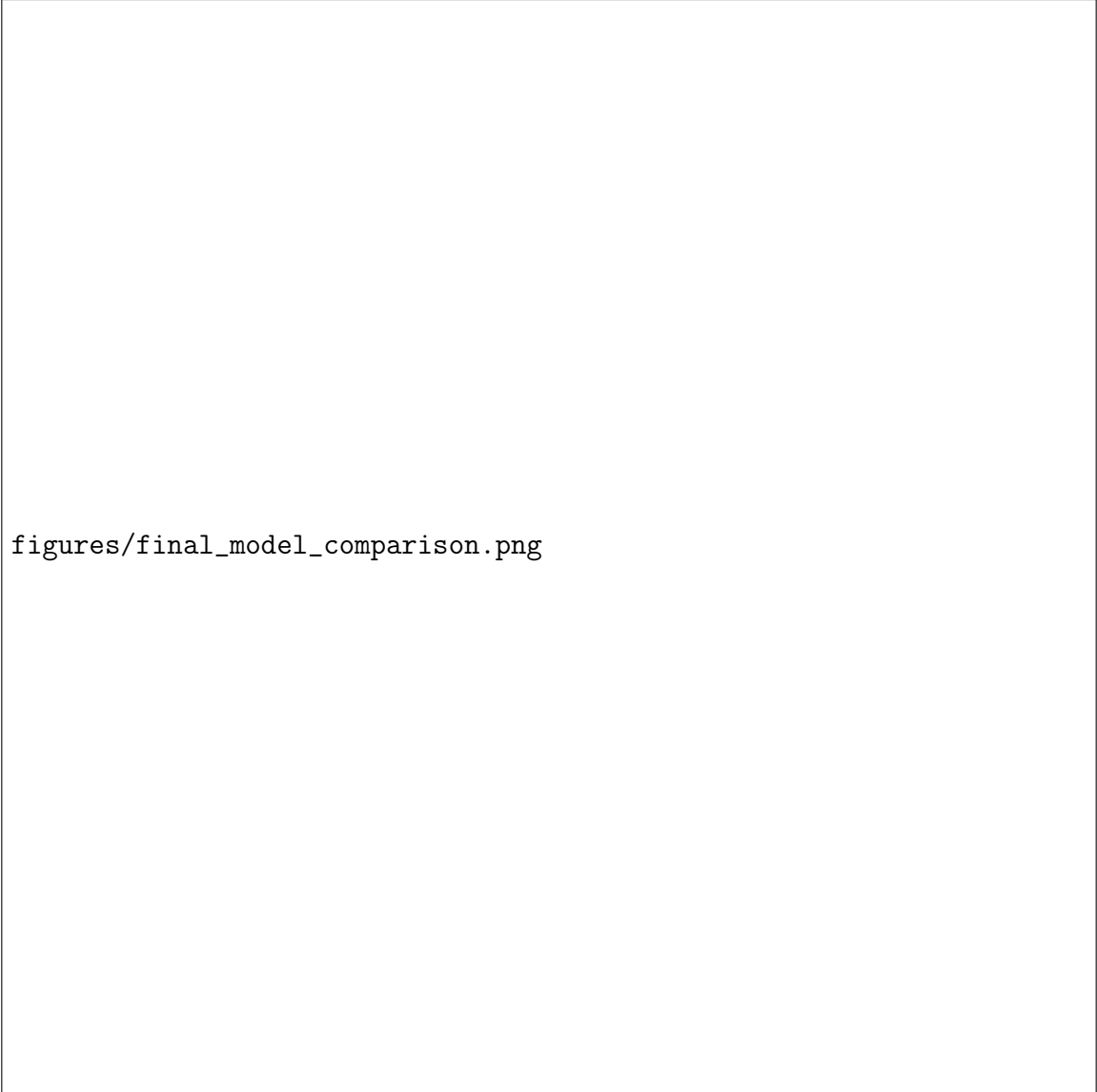
- **CO₂ emissions** are highly predictable ($R^2 > 0.98$) across all tree-based models, confirming strong deterministic relationships with socioeconomic scale variables.
- **Temperature anomalies** yield negative R^2 across all architectures, indicating that socioeconomic features alone lack sufficient signal to exceed a naive mean baseline.
- **Decision Tree baseline** achieves reasonable CO₂ performance ($R^2 = 0.982$) with minimal training time (0.15s), demonstrating that even simple non-linear models capture emission dynamics better than linear approaches.
- **Hyperparameter tuning** improved macro R^2 from 0.311 (default XGBoost) to 0.477 (tuned) primarily by stabilizing generalization, not by improving Temperature prediction.
- **Computational efficiency:** XGBoost/LightGBM achieve near-identical accuracy to Random Forest in <1 second vs. 142 seconds.

Interpretation Results (SHAP Analysis)

9.3.1 Global Feature Importance

Interpretation:

- **CO₂ drivers:** Population, GDP_per_Capita, and Energy_Consumption_Per_Capita dominate, aligning with established emission accounting frameworks (Kaya identity).
- **Temperature drivers:** Urbanization, Policy_Score, and Decade show the strongest (though still weak) associations, suggesting socioeconomic factors influence temperature only indirectly via development pathways.



figures/final_model_comparison.png

Figure 3: Macro-averaged model performance...

9.3.2 Marginal Effect Visualization

Pattern insights:

- **Urbanization** → **Temperature**: Non-linear threshold effect—warming contribution emerges only at extreme urbanization levels (>90th percentile), consistent with urban heat island literature.
- **Population** → **CO₂**: Strong monotonic linear relationship across the entire distribution, confirming population as a fundamental scaling factor for aggregate emissions.



Figure 4: SHAP summary (beeswarm) plots showing global feature importance for both targets. Red = high feature value, Blue = low feature value. Rightward shift = positive contribution to prediction. *Note: Replace this placeholder with your actual SHAP summary figure from `outputs/figures/`.*

Table 2: Model characteristics and practical considerations.

Dimension	Ridge	Decision Tree	Random Forest	XGBo
Assumption	Linear + L2 reg.	Single tree, depth-limited	Non-linear, bagging	Gradi
CO₂ R²	0.27	0.982	0.996	
Temp R²	0.001	-0.020	-0.013	
Train Time	0.23 s	0.15 s	142 s	
Overfitting Risk	Low (high bias)	Moderate (constrained)	Moderate	Low (
Interpretability	High (coefficients)	Very High (rules)	Medium (permutation)	Medi

figures/eda_target_correlations_both.png

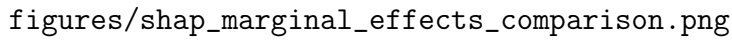
Figure 5: Top 10 features correlation...

COMPARATIVE ANALYSIS OF MODELS

Accuracy vs. Complexity Trade-offs

Key insights:

- (1) **Non-linearity is essential for CO₂:** Tree-based models capture interaction effects (e.g., $\text{GDP} \times \text{Energy_Usage}$) that linear models miss, explaining the R^2 jump from 0.27 (Ridge) to >0.98 (trees).
- (2) **Tuning stabilizes generalization:** Optuna optimization reduced test-set variance by selecting conservative hyperparameters (`max_depth=4`, `subsample=0.625`), improving macro R^2 by 53% without sacrificing CO₂ accuracy.
- (3) **Speed vs. accuracy:** XGBoost/LightGBM match Random Forest performance in



figures/shap_marginal_effects_comparison.png

Figure 6: Marginal effect plots (decile-binned SHAP averages) for key interpretable drivers. Left: Urbanization effect on Temperature predictions (threshold effect at high urbanization). Right: Population effect on CO₂ predictions (near-linear scaling). *Note: Replace with your actual marginal effects figure from `outputs/figures/`.*

<1 second, making them preferable for iterative experimentation and hyperparameter search.

- (4) **Target asymmetry:** No model overcomes Temperature’s weak signal, proving the limitation is data-driven (missing physical climate variables), not algorithmic.

Target-Specific Model Behavior

10.2.1 CO₂ Emissions: Deterministic Predictability

All tree-based models converge on near-perfect prediction ($R^2 > 0.98$), reflecting strong mechanistic links between socioeconomic scale and aggregate emissions. Performance

plateaus early—additional model complexity yields diminishing returns. SHAP analysis confirms that `Population` and `GDP_per_Capita` dominate predictions, consistent with the Kaya identity decomposition:

$$CO_2 = Population \times \frac{GDP}{Population} \times \frac{Energy}{GDP} \times \frac{CO_2}{Energy}$$

10.2.2 Temperature Anomaly: Fundamental Predictability Limit

Negative R^2 across all architectures indicates that socioeconomic features alone cannot reliably forecast temperature anomalies. This is not a modeling failure but a scientific finding: temperature dynamics depend on physical processes (ocean circulation, radiative forcing, aerosol effects) not captured in socioeconomic indicators. The consistent RMSE $\approx 0.66^\circ\text{C}$ across models suggests this represents the irreducible error floor for socioeconomic-only forecasting.

Practical Recommendations

- **For emission forecasting:** Use XGBoost or LightGBM with importance-based feature selection. The near-perfect performance validates socioeconomic indicators as reliable proxies for macro-level emission trajectories.
- **For temperature modeling:** Socioeconomic-only pipelines are insufficient. Future work should integrate physical climate variables (e.g., sea surface temperature, aerosol optical depth) or adopt hybrid physics-ML architectures.
- **For policy interpretation:** SHAP dependence plots reveal that urbanization and policy metrics matter most at extreme deciles. This suggests targeted interventions (e.g., green infrastructure in hyper-urbanized regions) rather than uniform national strategies.

REFERENCES

APPENDIX: PROJECT TIMELINE

Phase	Duration	Deliverables
Data Collection & Cleaning	Weeks 1–2	Cleaned panel dataset, EDA report
Feature Engineering	Weeks 3–4	Engineered feature set, selection report
Model Development	Weeks 5–7	Trained models, hyperparameter logs
Evaluation & Interpretation	Weeks 8–9	Performance tables, SHAP figures
Report Writing & Submission	Weeks 10–12	Final paper, code repository, presentation

Table 3: Proposed project timeline